# Casey Barton

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## PROFILE

I build the systems underneath agent deployments: the agent runtimes, tool and MCP integrations, eval harnesses, and observability that turn one-off field wins into reusable platform. Every agent I ship runs under an eval discipline first — golden sets, behavioral assertions, trace logging, judge criteria — and every production failure mode I have found has something built to catch it. Fail-closed by default.

Three years as a founder running an evidence-grade agent control plane from Honolulu: hand-rolled multi-agent loops with planning and tool-use, retrieval, memory, guardrails, and human handoffs; **65** MCP connectors exposing governed context to any MCP-compatible system; full run traces so any execution can be reconstructed; model-agnostic abstractions with benchmark loops that track cost, latency, and accuracy on every release.

## CORE COMPETENCIES

**Engineering:** TypeScript · Python · Node.js · SQL · Bash · REST APIs · containers · GitHub Actions (CI/CD) · Supabase · Postgres · event-driven systems

**AI & Evaluation:** agent runtimes · MCP servers & connectors · LLM evaluation harnesses · judge-in-the-loop patterns · trace intelligence · failure-mode instrumentation · benchmarking & experiments · memory systems · RAG · function calling · structured outputs · hand-rolled multi-agent loops · guardrails · fail-closed recovery

**Production:** production platform engineering · observability & run logging · latency & token-cost benchmarks · CI regression gates · decision audit receipts · documentation engineers actually read

## EXPERIENCE

**Founder / Applied AI Systems Architect** 2023 – Present

*GlacierEQ — Honolulu, HI*

- Own the harness: eval suites, traces, and judge-in-the-loop patterns that tell whether an agent is actually doing the job, against a baseline. Prompt and configuration changes are validated by eval harnesses before anything touches production — an agent that cannot be measured cannot be trusted.
- Design agent systems for failure: instrument the failure modes found in production — stuck loops, bad retrievals, tool misuse, context blowouts — and build the catchers: replay loops, fail-closed gates, behavioral assertions, and recovery restricted to evidence declared before reassessment.
- Author **65** MCP connectors and integrations that expose tools and structured resources to any MCP-compatible system — the shared, governed context layer: ERPs, CRMs, data pipelines, webhooks — with evidence boundaries and audit receipts.
- Build model-agnostic: abstract the model layer across providers and benchmark when a new release lands — agent primitives tested against datasets, per-task latency and token-cost instrumentation, findings turned into configuration improvements.
- Run an evidence-grade agent control plane across a **1,413-repository** estate: planning and tool-use loops, retrieval over a domain ontology, memory, guardrails, typed handoff contracts, human

approval gates on consequential actions — backed by a **989-test** suite, **95/95** verification checks green.

- Turn field learning into reusable infrastructure: one-off fixes become shared patterns with written design docs, versioned eval suites, and the same governed context for every downstream tool.

## EDUCATION

**University of Hawaii at Manoa** 2016
*Bachelor of Science*

**AWS Cloud Institute** 2025
*Cloud Computing program*

**Colorado State University**
*Undergraduate coursework*

## SELECTED PROOF

**Job-App Helix — evaluation-gated campaign engine**

- Attributable campaign engine with immutable evidence models, stable finding codes, and fail-closed **GO / NO-GO** decisions; recovery restricted to contingency evidence declared before reassessment.
- Verification gates on every agent action: no agent completion claim without independent provider readback — the eval gate decides, not the agent.

**GlacierEQ estate — production agent systems**

- Durable patterns for claim tracking, evidence boundaries, audit receipts, and human-in-the-loop control across the full estate.
- Per-task latency instrumentation and decision audit logs across agent runs — the tracing practice that makes benchmarking honest.

## SKILLS

**Languages:** TypeScript · Python · SQL · Bash · JavaScript / Node.js

**AI / Evaluation:** agent runtimes · MCP servers & connectors · eval harnesses · judge-in-the-loop · trace intelligence · failure-mode instrumentation · benchmarking · memory systems · guardrails · typed handoff contracts

**Production:** platform engineering · observability & run logging · CI regression gates · fail-closed recovery · cost/latency tradeoffs · documentation & handover

*Evidence boundary: portfolio repositories are independent work. Domain-inspired demonstrations are not claims of company affiliation, production deployment, or certification.*